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Matriculation No.: A0255566U

Fumehood No: 15

Lab Group: Thursday

Dates of Experiment: 22/08 & 29/08 & 05/09

CM3191 (AY2024/25 Semester 1)

INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

This lab proforma must be handwritten and its softcopy must be submitted together with the appendices (in ONE file) including the 1H-NMR, IR and ESI-MS spectra, of your products obtained, and your own datasheet(s) for all lab sessions, via Canvas-Assignments by 10:00 a.m. on 8-Nov-2024 (Fri.; week 12).

1. State the balanced reaction equation for the synthesis of $RuCl_2(PPh_3)_3$ in Session I.

$RuCl_3 \cdot xH_2O + 4PPh_3 \rightarrow RuCl_2(PPh_3)_3 + HCl + (x-1)H_2O + OPPh_3$

2. Calculate the % yield of your $RuCl_2(PPh_3)_3$ obtained in Session I. [Note: Detailed calculation steps are required.]

Moles of PPh_3 used : $0.4997\ (g) \div 262.29\ (g/mol) = 1.90514\ mmol$

Moles of $RuCl_3 \cdot xH_2O$ used (with anhydrous $RuCl_3$ molecular weight)

: $0.0812\ (g) \div 207.4\ (g/mol) = 0.39146\ mmol$

Since $RuCl_3 \cdot xH_2O$ is the limiting reagent, the theoretical yield of $RuCl_2(PPh_3)_3$ is

: $0.39146\ (mmol) \times 958.83\ (g/mol) \times 10^{-3} = 0.37534\ g$

Since the mass of dried $RuCl_2(PPh_3)_3$ is 0.1961g, the percentage yield is

: $\frac{0.1961\ (g)}{0.37534\ (g)} \times 100\% = 52.25\%$ (4 s.f.)

3. Use no more than 50 words to interpret the ESI-MS signals obtained in the Zoom scan spectrum for your $RuCl_2(PPh_3)_3$ in Session I.

[Notes: (i) Do NOT tabulate your experimental results. (ii) Only the first 50 words will be marked if more than 50-word answer is given.]

Three clusters of peaks are observed where highest intensity peak contains the most abundant ^{102}Ru (31.6%.)[1] and ^{35}Cl (75.53%.)[2]. Combinations of less abundant isotopes give 11 less intense peaks. 660.92 base peak is $[RuCl(PPh_3)_2]^+$ with one PPh_3 and Cl fragmented from $RuCl_2(PPh_3)_3$. 625.00 and 701.68 indicate $[Ru(PPh_3)_2H]^+$ and $[RuCl(CH_3CN)(PPh_3)_2]^+$ respectively.

word count : 50

1. Bermingham, K. R.; Walker, R. J., *EPSL*. 2017, 474, 466-473
2. Meyerson, S., *Anal. Chem.* 1961, 33(7), 964.

4. Tabulate and assign the IR signals obtained for your $RuCl_2(PPh_3)_3$ in Session I.

Wavenumber (cm^{-1})	Intensity	Vibrational mode assignment[3]
3047.90	medium & broad	C-H aromatic stretch (a_1)
2360.13	weak & broad	atmospheric CO_2
1569.30	weak & broad	phenyl ring overtone
1479.38	strong & sharp	C=C aromatic stretch (a_1)
1432.40	strong & sharp	C=C aromatic stretch (b_2)
1316.23	weak & broad	C-C aromatic stretch (B_2)
1186.52	weak & sharp	C-H aromatic bending (a_1)
1086.07	strong & sharp	P-C stretching (a_1)
997.80	weak & sharp	phenyl ring in-phase stretching (a_1)
741.93	strong & sharp	C-H aromatic bending (b_1)
649.49	strong & sharp	P-C stretching (a_1)
542.93	medium & sharp	CCC bending (b_2)
518.51	strong & sharp	P-C bending (b_1)

3. Deacon, G. B.; Green, J. H. S., *Spectrochim Acta A Mol Biomol Spectrosc.* 1968, 24(7), 845-852

5. Use no more than **50** words to comment the ^{1}H-NMR signals obtained in Session II (controlled study).

[Notes: (i) Do NOT tabulate your experimental results. (ii) Only the first 50 words will be marked if more than 50-word answer is given.]

Two intense singlet 6.94 and 2.42 ppm peaks indicate mesitylene and small singlet 2.68 ppm peak indicate acetophenone. Multiplet 7.47–7.45 and 7.39–7.37 ppm for aromatic proton, quartet at 4.98-4.93 ppm, and doublet at 1.60 ppm belong to 1-phenylethanol. Absence of 2-propanol peaks implies reaction quenching before 1-phenylethanol conversion to 2-propanol.

word count: 50
(taking the peak range (e.g. 7.47 -7.45) as 1 word according to Microsoft Word's system)

6. Calculate the % yield of acetophenone obtained in your Session II. [Note: Detailed calculation steps are required.]

The integrated value of the peak is 0.12 for 2.68 ppm and 9.00 for 2.44 ppm which belong to acetophenone and mesitylene, respectively. Hence, $\frac{n_{acetophenone}}{n_{mesitylene}} = \frac{\frac{0.12}{3}}{\frac{9}{9}} = 0.04$

moles of mesitylene used: 0.07 (mL) × 0.8637 (g/mL) ÷ 120.14 (g/mol) = 0.50503 mmol

$n_{acetophenone}$ = 0.04 × 0.50303 = 0.020121 mmol

moles of 1-phenylethanol used: 0.06 (mL) × 1.013 (g/mL) ÷ 122.17 (g/mol) = 0.49752 mmol

Thus, the percentage yield is:

$\frac{0.020121 \text{ (mmol)}}{0.49752 \text{ (mmol)}} \times 100\% = 4.044\%$ (4 s.f.)

∴ 4.044 %.

7. Use no more than **50** words to comment the ^{1}H-NMR signals obtained in Session III (study of catalytic reaction).

[Notes: (i) Do NOT tabulate your experimental results. (ii) Only the first 50 words will be marked if more than 50-word answer is given.]

Alongside 2.68 ppm peak intensity being tripled, acetophenone's ortho-, para-, and meta-peaks appeared at 7.79–7.73, 7.68–7.62 and 7.57–7.54 ppm, respectively. 1.44 ppm doublet peak indicates 2-propanol, another reaction product. These peaks show that yield is higher with $RuCl_2(PPh_3)_3$, matching with the observed green filtrate indicating oxidised catalyst.

word count: 47

8. Calculate the % yield of acetophenone obtained in your Session III. [Note: Detailed calculation steps are required.]

The integrated value of the peak is 0.28 for 2.68 ppm and 9.00 for 2.42 ppm which belongs to acetophenone and mesitylene, respectively.

Thus, $\frac{n_{acetophenone}}{n_{mesitylene}} = \frac{\frac{0.28}{3}}{\frac{9.00}{9}} \approx 0.093333$

As 0.50303 mmol of mesitylene was added to the ^{1}H NMR sample (calculation in Question 6),

$n_{acetophenone} = 0.093333 \times 0.50303\ (mmol) = 0.046949\ mmol$

Therefore, the percentage yield is:

$\frac{0.046946\ (mmol)}{0.44752\ (mmol)} \times 100\% = 9.437\%$ (4 s.f.)

∴ 9.437%

9. Calculate the TOF of the $RuCl_2(PPh_3)_3$ in the catalytic reaction studied. [Note: Detailed calculation steps are required.]

Moles of $RuCl_2(PPh_3)_3$ used: $0.0619\ (g) \div 958.83\ (g/mol) = 6.4558 \times 10^{-5}\ mol$

From Question 8, the moles of acetophenone produced is 0.046949 mmol
$= 4.6949 \times 10^{-5}\ mol$

$$TOF = \frac{\frac{(\text{amount of product})}{(\text{amount of catalyst})}}{\text{time}} = \frac{\frac{4.6949 \times 10^{-5}}{6.4558 \times 10^{-5}}}{60} = 12.12 \times 10^{-3}\ min^{-1}$$

$\therefore 12.12 \times 10^{-3}\ min^{-1}$

10. Use no more than **30** words to comment the value of TOF obtained in Question 9.

[Note: Only the first 30 words will be marked if more than 30-word answer is given.]

TOF is lower than the literature value by order of 1000, suggesting $RuCl_2(PPh_3)_3$ is highly inactive [4]. Recrystallisation to increase catalyst purity and performing reaction in anhydrous condition will help increase TOF.

word count: 30

4. Almeida, L.S.; Beller, M.; Wang, G.; Bäckvall, J., *Chem. Eur. J.* **1996**, 2(12), 1533–1536

END OF LAB PROFORMA

Appendix

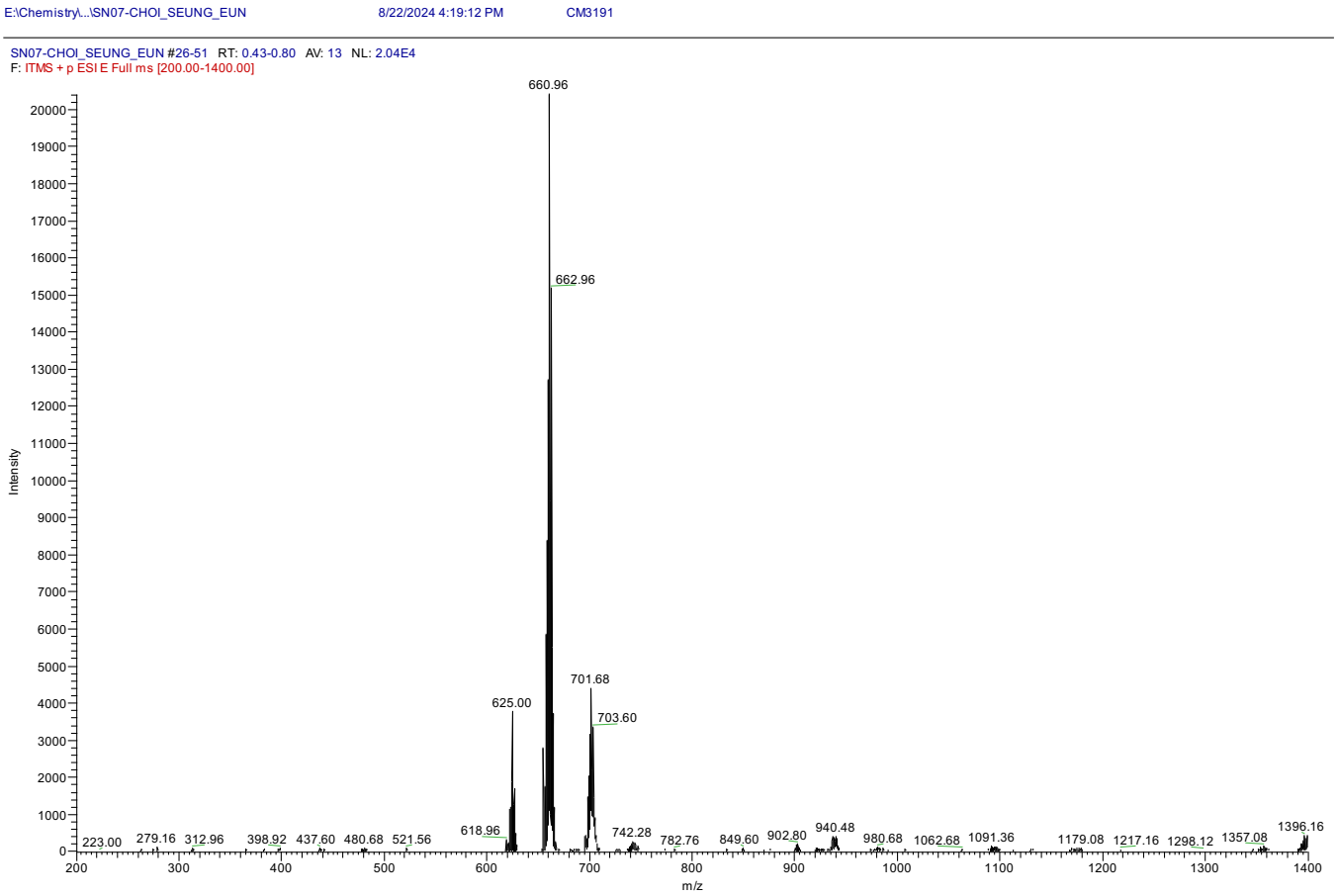


Figure 1: ESI-MS full spectrum for $RuCl_2(PPh_3)_3$ catalyst

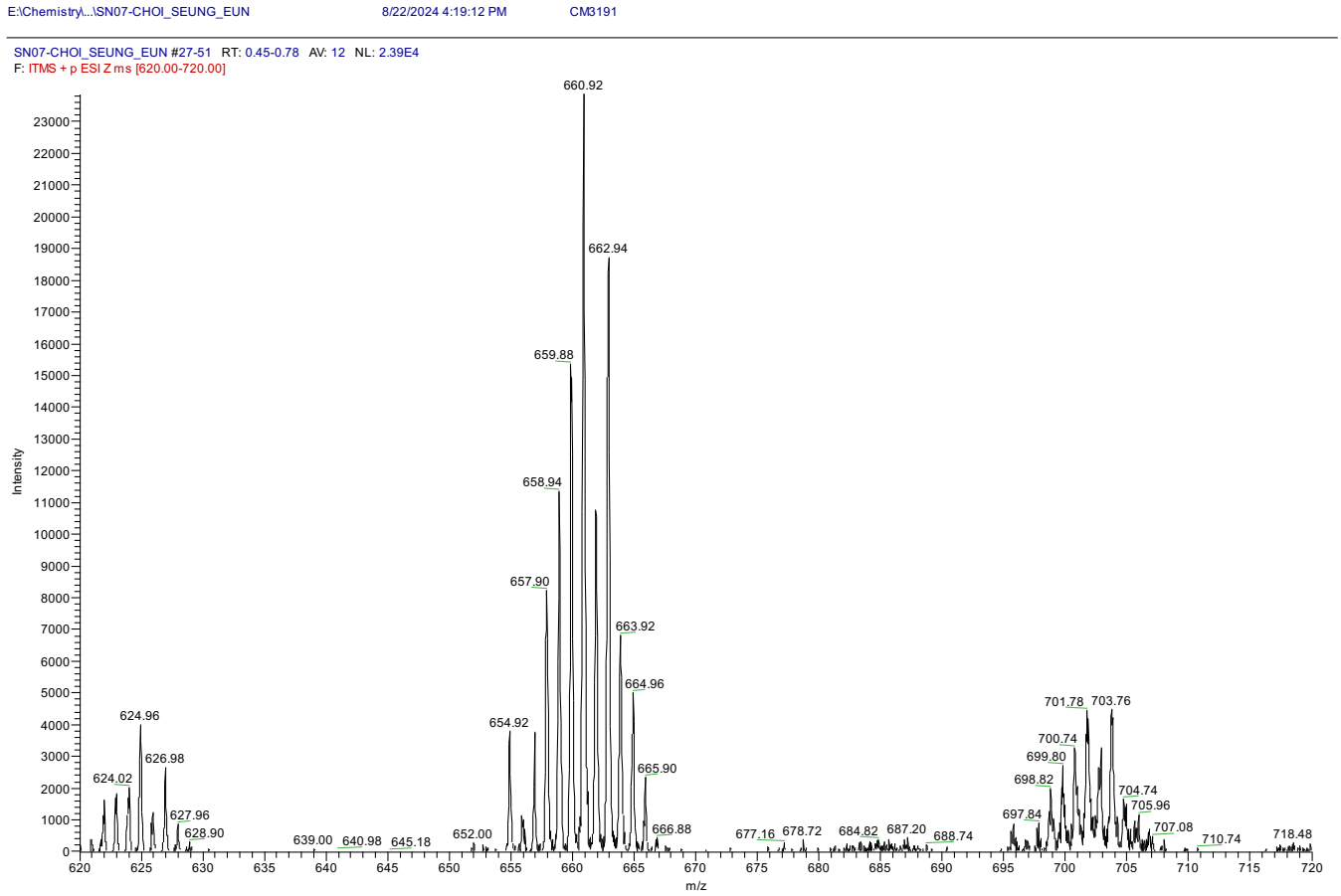


Figure 2: ESI-MS zoomed spectrum for $RuCl_2(PPh_3)_3$ catalyst

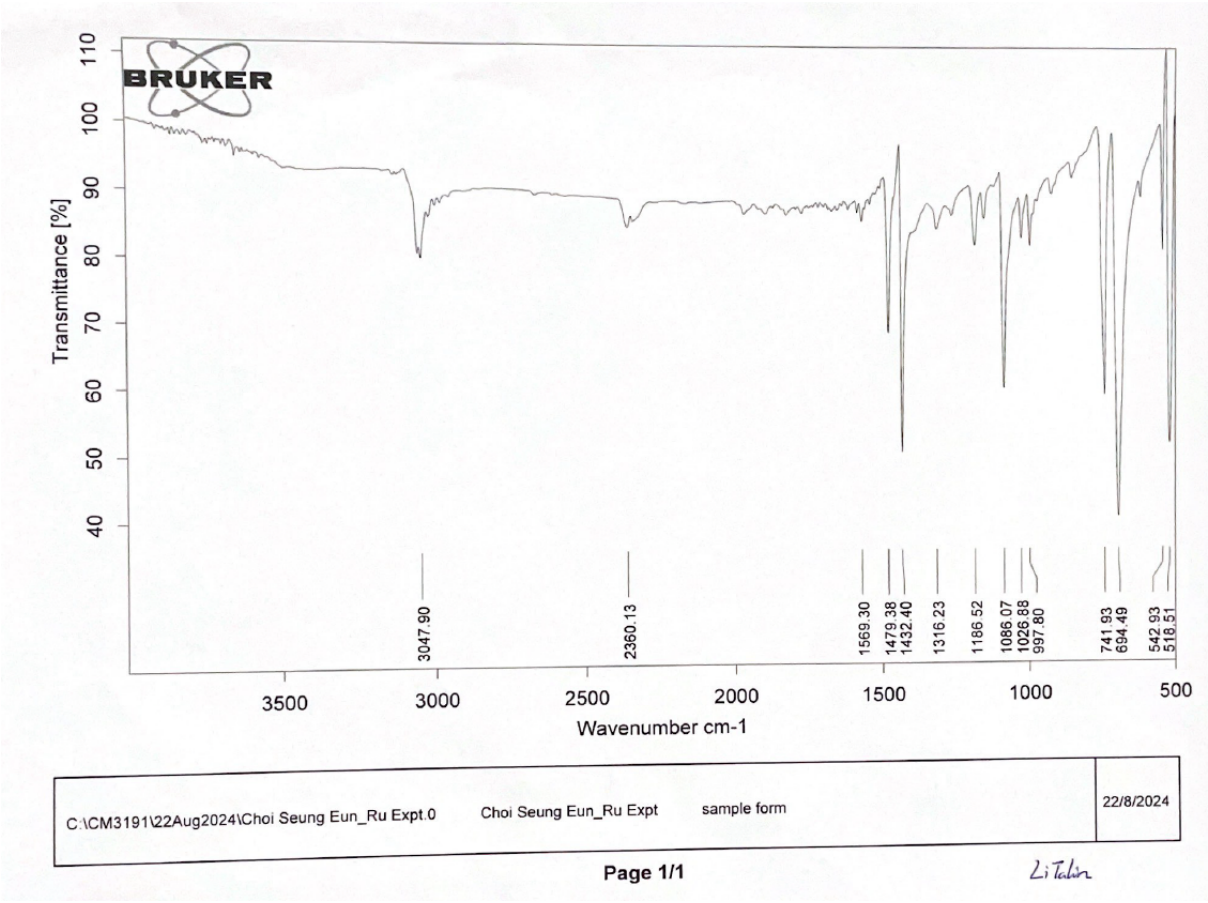


Figure 3: KBr IR spectrum for $RuCl_2(PPh_3)_3$ catalyst

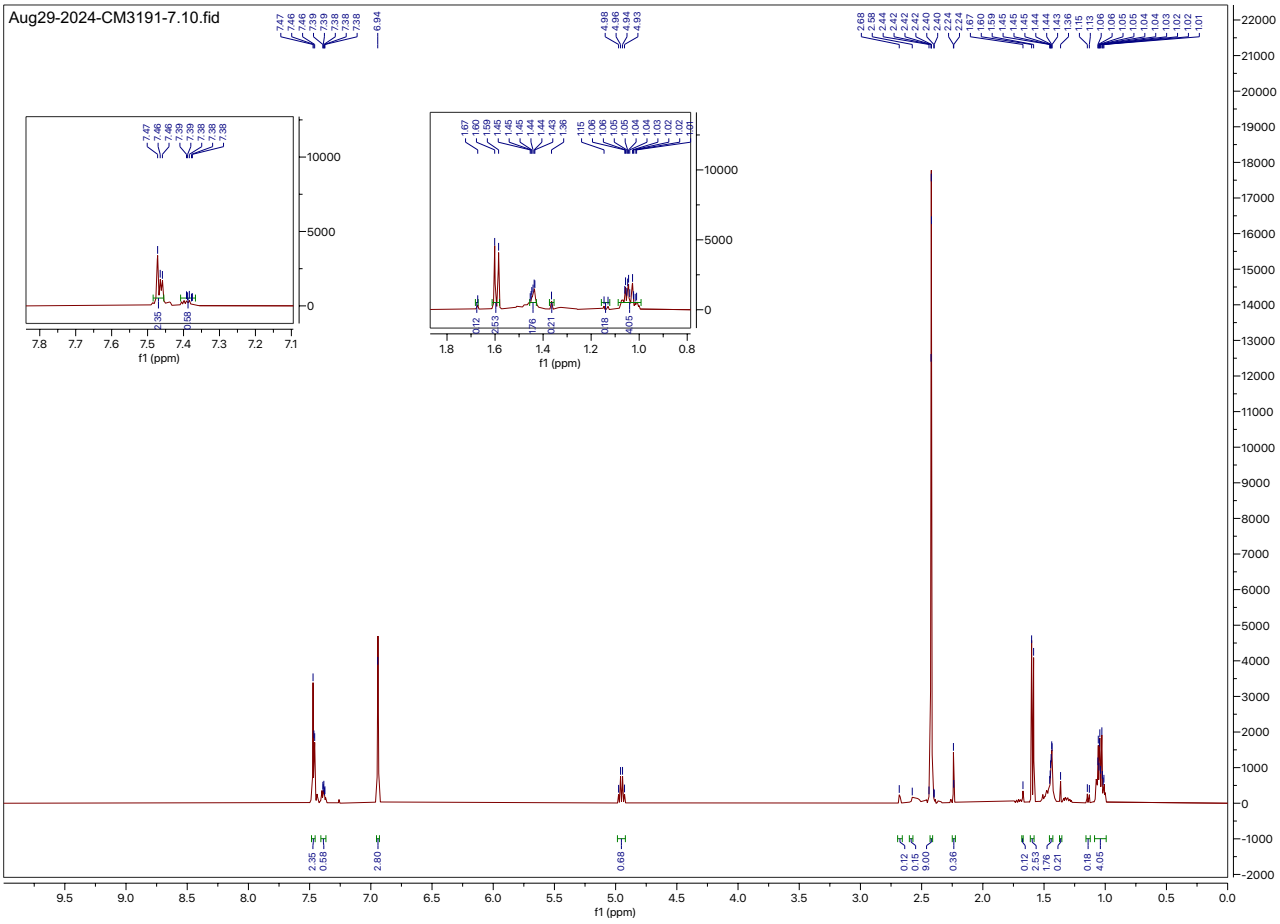


Figure 4: ^{1}H NMR spectrum of Session II experiment product

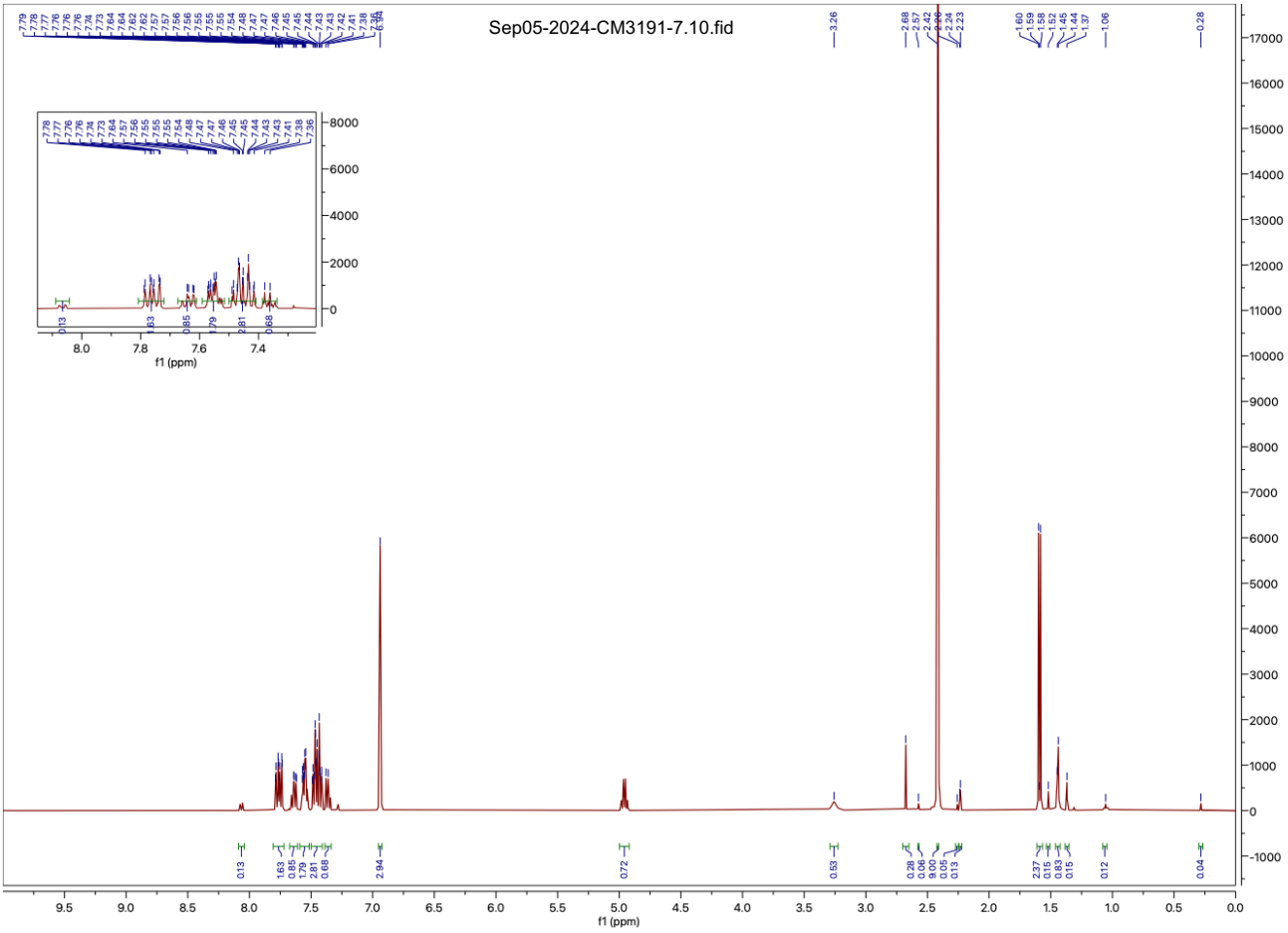


Figure 5: ^{1}H NMR spectrum of Session III experiment product

Data Sheet – Inorganic Experiment 1 (Session 1)

22/08/2024 (Thursday) / FC : 15 / Choi Seung Eun

1. Mass of PPh_3 : 0.4997g
2. Mass of $RuCl_3 \cdot xH_2O$: 0.0812g
3. Exact reflux time : 95 min (11:15 ~12:50/oil bath temperature : 90 °C) + setting : 150°C
4. Mass of dried $RuCl_2(PPh_3)_3$ obtained : 1.3174 − 1.1213 = 0.1961g
5. IR characterisation method : ATR-IR & KBr disc
6. Observations
   - $RuCl_3 \cdot xH_2O$ solution is brownish-black ↔ PPh_3 solution is transparent
   - after reflux, black powder was formed & the solution became transparent again (yellow colour)
7. Notes
   - when transferring $RuCl_3 \cdot xH_2O$ solution to PPh_3 solution, have stirring going on (reaction starts right after combining the two chemicals)
   - don't damp rbf and drying tube + don't put cap on reaction mixture. instead, place the drying tube while leaving the rbf while setting up the condensor (cap will pop off)
   - no need to exactly weigh the chemicals – if slow, react with H_2O
   - don't change setting temperature after reflux started (unless temperature goes up too high)
   - cool down rbf in ice bath (before that, wipe oil using hexane & tissue)

TA signature: ZiTaZn

Data Sheet - Inorganic Experiment 1 (Session 2)

29/08/2024 (Thursday) / FC : 15 / Choi Seung Eun

1. Mass of K_2CO_3 : 0.0711 g

2. Reflux condition : 60 min (10:26 ~ 11:26), oil bath temperature : 62 °C (setting : 90 °C)

3. ~~Mass of (impure) product :~~

4. Observations
- 1-phenylethanol is a clear solution & K_2CO_3 is white powder
- filtrate also transparent
- after rotavap, only few droplets of product (filtrate) remaining on the sides of rbf
- vacuum to gas adapter : 15 min (12:11 ~ 12:26)

5. Notes

A signature : Choi Talon

Data Sheet - Inorganic Experiment 1 (Session 3)

05/09/2024 (Thu) / 조 : 15 / Choi Seung Eun

1. Mass of K_2CO_3 : 0.0712 g

2. Mass of $RuCl_2(PPh_3)_3$: 0.0619 g

3. Reflux condition : 60 min (10 : 29 ~ 11 : 29) / temperature : ~~64~~ 70 °C (setting : 100°C)

4. Notes & Observations
- solution is greenish black after adding all starting materials → after reflux, dark green solution → opaque dark green when hexane added to quench the reaction
- green filtrate & greenish black powder on celite.
- after first filtration, green solid was observed in the filtrated (due to residual dichloromethane) hence, filtration was performed again. → mostly clear (colourless) with faint green colour

~~-100.01g~~

- after concentration under vacuum, dark olive green solution

Li Tolun

TA signature :